TRAIN

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THE TRAIN

- Over the years, the exam pattern has changed with the increase in the number of applicants and every year candidates notice a new pattern or format in which questions are asked for various topics in the syllabus.
- It is important that a candidate is aware of the types in which a question may be framed or asked in the examination to avoid any risk of losing marks.
- Thus, given below are the type of questions which may be asked from the train-based problems:
- **Time Taken by Train to Cross any stationary Body or Platform –** Question may be asked where the candidate has to calculate the time taken by a train to cross a stationary body like a pole or a standing man or a platform/ bridge
- **Time Taken by 2 trains to cross each other –** Another question that may be asked is the time two trains might take to cross each other
- **Train Problems based on Equations –** Two cases may be given in the question and the candidates will have to form equations based on the condition given
- Aspirants must also refer to the important pointer given below for the Train problems.
- The image given below mentions the basic points a candidate must remember in order to answer the train based word problems:

THE TRAIN

- **Speed of the Train = Total distance covered by the train / Time taken**
- If the length of two trains is given, say a and b, and the trains are moving in **opposite directions** with speeds of x and y respectively, then the **time taken by trains to cross each other = {(a+b) / (x+y)}**
- If the length of two trains is given, say a and b, and they are moving in the **same direction**, with speeds x and y respectively, then the **time is taken to cross each other = {(a+b) / (x-y)}**
- When the **starting time of two trains is the same from x and y towards each other** and after crossing each other, they took t1 and t2 time in reaching y and x respectively, then the **ratio between the speed of two trains = √t2 : √t1**
- If two trains leave x and y stations at time t1 and t2 respectively and travel with speed L and M respectively, then distanced from x, where two trains meet is = (t2 – t1) × {(product of speed) / (difference in speed)}
- The average speed of a train without any stoppage is x, and with the stoppage, it covers the same distance at an average speed of y, then **Rest Time per hour = (Difference in average speed) / (Speed without stoppage)**
- If two trains of equal lengths and different speeds take t1 and t2 time to cross a pole, then the **time taken by them to cross each other if the train is moving in opposite direction = (2×t1×t2) / (t2+t1)**
- If two trains of equal lengths and different speeds take t1 and t2 time to cross a pole, then the **time taken by them to cross each other if the train is moving in the same direction = (2×t1×t2) / (t2-t1)**

THE TRAIN

Q 1. A train 110 m in length runs through a station at the rate of 36 km per hour. How long will it take to pass a given point?

(1) 11 sec.

(2) 12 sec.

(3) 13 sec.

(4) 15 sec.

(5) None of these

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Q 1. A train 110 m in length runs through a station at the rate of 36 km per hour. How long will it take to pass a given point?

(1) 11 sec.

(2) 12 sec.

(3) 13 sec.

(4) 15 sec.

(5) None of these

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THE TRAIN

Q 2. A train 540 m long is running with a speed of 72 km/hr. In what time will it pass a tunnel 160 m long?

(1) 40 sec.

(2) 30 sec.

(3) 35 sec.

(4) 42 sec.

(5) None of these

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THE TRAIN

Q 2. A train 540 m long is running with a speed of 72 km/hr. In what time will it pass a tunnel 160 m long?

(1) 40 sec.

(2) 30 sec.

(3) 35 sec.

(4) 42 sec.

(5) None of these

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THE TRAIN

Q 3. Two trains 70 m and 80 m long respectively, run at the rates of 68 and 40 km an hour respectively on parallel rails in opposite directions. How long do they take to pass each other?

(1) 5 sec.

(2) 10 sec.

(3) 12 sec.

(4) 6 sec.

(5) None of these

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THE TRAIN

Q 3. Two trains 70 m and 80 m long respectively, run at the rates of 68 and 40 km an hour respectively on parallel rails in opposite directions. How long do they take to pass each other?

(1) 5 sec.

(2) 10 sec.

(3) 12 sec.

(4) 6 sec.

(5) None of these

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THE TRAIN

Q 4. A train 110 metres long travels at 60 km/hr. How long does it take to cross another train 170 metres long, running at 54 km/hr. in the same direction?

(1) 2 min 40 sec.

(2) 2 min 48 sec.

(3) 3 min 48 sec.

(4) 3 min 40 sec.

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THE TRAIN

Q 4. A train 110 metres long travels at 60 km/hr. How long does it take to cross another train 170 metres long, running at 54 km/hr. in the same direction?

(1) 2 min 40 sec.

(2) 2 min 48 sec.

(3) 3 min 48 sec.

(4) 3 min 40 sec.

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THE TRAIN

Q 5. Two trains travel in the same direction at 56 km and 29 km an hour and the faster train passes a man in the slower train in 16 seconds. Find the length of the faster train.

(1) 100 m

(2) 120 m

(3) 124 m

(4) Data inadequate

(5) None of these

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THE TRAIN

Q 5. Two trains travel in the same direction at 56 km and 29 km an hour and the faster train passes a man in the slower train in 16 seconds. Find the length of the faster train.

(1) 100 m

(2) 120 m

(3) 124 m

(4) Data inadequate

(5) None of these

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THE TRAIN

Q 6. A train running at 24 km/hr takes 30 seconds to pass a platform. Next, it takes 10 seconds to pass a man walking at 12 km/hr in the opposite direction. Find the length of the train.

(1) 50 m

(2) 100 m

(3) 75 m

(4) 120 m

(5) None of these

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THE TRAIN

Q 6. A train running at 24 km/hr takes 30 seconds to pass a platform. Next, it takes 10 seconds to pass a man walking at 12 km/hr in the opposite direction. Find the length of the train.

(1) 50 m

(2) 100 m

(3) 75 m

(4) 120 m

(5) None of these

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THE TRAIN

Q 7. A train running at 25 km/hr. takes 18 seconds to pass a platform. Next it takes 10 seconds to pass a man walking at the rate of 7 km/hr. in the same direction. Find the length of the platform and the length of the train.

(1) 25m, 50m

(2) 45m, 85m

(3) 75m, 50m

(4) 50m, 80m

(5) None of these

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THE TRAIN

Q 7. A train running at 25 km/hr. takes 18 seconds to pass a platform. Next it takes 10 seconds to pass a man walking at the rate of 7 km/hr. in the same direction. Find the length of the platform and the length of the train.

(1) 25m, 50m

(2) 45m, 85m

(3) 75m, 50m

(4) 50m, 80m

(5) None of these

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THE TRAIN

Q 8. 60 metres long train crosses a tunnel of length 40 metres in 10 seconds. Find the time for train to cross a man standing on a platform of length 65 metres.

(1) 6 sec.

(2) 8 sec.

(3) 5 sec.

(4) 4 sec.

(5) None of these these

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THE TRAIN

Q 8. 60 metres long train crosses a tunnel of length 40 metres in 10 seconds. Find the time for train to cross a man standing on a platform of length 65 metres.

(1) 6 sec.

(2) 8 sec.

(3) 5 sec.

(4) 4 sec.

(5) None of these these

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THE TRAIN

Q 9. Two trains start at the same time from Patna and Gaya and proceed towards each other at the rate of 60 km and 40 km per hour respectively. When they meet, it is found that one train has travelled 20 km more than the other. Find the distance between Gaya and Patna.

(1) 100 km

(2) 80 km

(3) 120 km

(4) 90 km

(5) None of these

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THE TRAIN

Q 9. Two trains start at the same time from Patna and Gaya and proceed towards each other at the rate of 60 km and 40 km per hour respectively. When they meet, it is found that one train has travelled 20 km more than the other. Find the distance between Gaya and Patna.

(1) 100 km

(2) 80 km

(3) 120 km

(4) 90 km

(5) None of these

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THE TRAIN

Q 10. Two stations A and B are 110 kms apart on a straight line. One train starts from A at 7 AM and travels towards B at 20 km/hr speed. Another train starts from B at 8 AM and travels towards A at 25 km/hr speed. At what time will they meet?

(1) 9 AM

(2) 10 AM

(3) 11 AM

(4) 11.5 AM

(5) None of these

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THE TRAIN

Q 10. Two stations A and B are 110 kms apart on a straight line. One train starts from A at 7 AM and travels towards B at 20 km/hr speed. Another train starts from B at 8 AM and travels towards A at 25 km/hr speed. At what time will they meet?

(1) 9 AM

(2) 10 AM

(3) 11 AM

(4) 11.5 AM

(5) None of these

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THE TRAIN

Q 11. Two stations A and B are 60 km apart on a straight line. A train starts from A towards B at the rate of 20 km/hr. 3 hours later another train starts from B and travels towards A at the rate of 25 km/hr. When will the first train meet to the second train?

(1) 1 hr.

(2) 2 hrs.

(3) 3 hrs.

(4) 4 hrs.

(5) None of these

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THE TRAIN

Q 11. Two stations A and B are 60 km apart on a straight line. A train starts from A towards B at the rate of 20 km/hr. 3 hours later another train starts from B and travels towards A at the rate of 25 km/hr. When will the first train meet to the second train?

(1) 1 hr.

(2) 2 hrs.

(3) 3 hrs.

(4) 4 hrs.

(5) None of these

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THE TRAIN

Q 12. Two stations A and B are 95 km apart on a straight line. A train starts from A and travels towards B at 35 km/hr. Another train, starting from B 30 minutes earlier, travels towards A at 40 km/hr. When will the first train meet to the second train?

(1) 2 hrs.

(2) 1 hr.

(3) 1.5 hrs.

(4) 2.5 hrs.

(5) None of these

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THE TRAIN

Q 12. Two stations A and B are 95 km apart on a straight line. A train starts from A and travels towards B at 35 km/hr. Another train, starting from B 30 minutes earlier, travels towards A at 40 km/hr. When will the first train meet to the second train?

(1) 2 hrs.

(2) 1 hr.

(3) 1.5 hrs.

(4) 2.5 hrs.

(5) None of these

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THE TRAIN

Q 13. Two trains of the same length but with different speeds pass a static pole in 5 seconds and 6 seconds respectively. In what time will they cross each other when they are moving in the same direction.

(1) 1 hr.

(2) 50 sec.

(3) 1 min

(4) 60 min

(5) None of these

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THE TRAIN

Q 13. Two trains of the same length but with different speeds pass a static pole in 5 seconds and 6 seconds respectively. In what time will they cross each other when they are moving in the same direction.

(1) 1 hr.

(2) 50 sec.

(3) 1 min

(4) 60 min

(5) None of these

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THE TRAIN

Q 14. Two trains of the same length but with different speeds pass a static pole in 10 seconds and 15 seconds respectively. In what time will they cross each other when they are moving in the opposite direction.

(1) 13 sec.

(2) 11 sec.

(3) 12.5 sec.

(4) 12 sec.

(5) None of these

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THE TRAIN

Q 14. Two trains of the same length but with different speeds pass a static pole in 10 seconds and 15 seconds respectively. In what time will they cross each other when they are moving in the opposite direction.

(1) 13 sec.

(2) 11 sec.

(3) 12.5 sec.

(4) 12 sec.

(5) None of these

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THE TRAIN

Q 15. Two trains of the length 200 m and 250 m respectively with different speeds pass a static pole in 8 seconds and 14 seconds respectively. In what time will they cross each other when they are moving in the same direction?

(1) 63 sec.

(2) 64 sec.

(3) 72 sec.

(4) 81 sec.

(5) None of these

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THE TRAIN

Q 15. Two trains of the length 200 m and 250 m respectively with different speeds pass a static pole in 8 seconds and 14 seconds respectively. In what time will they cross each other when they are moving in the same direction?

(1) 63 sec.

(2) 64 sec.

(3) 72 sec.

(4) 81 sec.

(5) None of these

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THE TRAIN

Q 16. Two trains running at the rates of 45 and 36 km an hour respectively, on parallel rails in opposite directions, are observed to pass each other in 8 seconds, and when they are running in the same direction at the same rate as before, a person sitting in the faster train observes that he passes the other in 30 seconds. Find the length of the trains.

(1) 105 m, 75m

(2) 50m, 25m

(3) 120m, 90m

(4) 100m, 75m

(5) None of these

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THE TRAIN

Q 16. Two trains running at the rates of 45 and 36 km an hour respectively, on parallel rails in opposite directions, are observed to pass each other in 8 seconds, and when they are running in the same direction at the same rate as before, a person sitting in the faster train observes that he passes the other in 30 seconds. Find the length of the trains.

(1) 105 m, 75m

(2) 50m, 25m

(3) 120m, 90m

(4) 100m, 75m

(5) None of these

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THE TRAIN

Q 17. A train 75 metres long overtook a man who was walking at the rate of 6 km/hr. and passed him in 18 seconds. Again, the train overtook a second person in 15 seconds. At what rate was the second person travelling?

(1) 3 km/hr.

(2) 5 km/hr.

(3) 8 km/hr.

(4) 9 km/hr.

(5) None of these

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THE TRAIN

Q 17. A train 75 metres long overtook a man who was walking at the rate of 6 km/hr. and passed him in 18 seconds. Again, the train overtook a second person in 15 seconds. At what rate was the second person travelling?

(1) 3 km/hr.

(2) 5 km/hr.

(3) 8 km/hr.

(4) 9 km/hr.

(5) None of these

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THE TRAIN

Q 18. A good train and a passenger train are running on parallel tracks in the same direction. The driver of the goods train observes that the passenger train coming from behind overtakes and crosses his train completely in 30 seconds. Whereas a passenger on the passenger train marks that he crosses the goods train in 20 seconds. If the speeds of the trains be in the ratio of 1 : 2, find the ratio of their lengths.

(1) 3 : 2

(2) 3 : 1

(3) 2 : 1

(4) 4 : 1

(5) None of these

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THE TRAIN

Q 18. A good train and a passenger train are running on parallel tracks in the same direction. The driver of the goods train observes that the passenger train coming from behind overtakes and crosses his train completely in 30 seconds. Whereas a passenger on the passenger train marks that he crosses the goods train in 20 seconds. If the speeds of the trains be in the ratio of 1 : 2, find the ratio of their lengths.

(1) 3 : 2

(2) 3 : 1

(3) 2 : 1

(4) 4 : 1

(5) None of these

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THE TRAIN

Q 19. A train covers a distance between stations A and B in 2 hours. If the speed is reduced by 6 km/hr, it will cover the same distance in 3 hours. What is the distance between the two stations A and B (in km)? Also, find the speed of the train.

(1) 36 km, 18 km/hr.

(2) 42 km, 21 km/hr

(3) 18 km, 9 km/hr

(4) 28 km, 14 km/hr.

(5) None of these

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THE TRAIN

Q 19. A train covers a distance between stations A and B in 2 hours. If the speed is reduced by 6 km/hr, it will cover the same distance in 3 hours. What is the distance between the two stations A and B (in km)? Also, find the speed of the train.

(1) 36 km, 18 km/hr.

(2) 42 km, 21 km/hr

(3) 18 km, 9 km/hr

(4) 28 km, 14 km/hr.

(5) None of these

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THE TRAIN

Q 20. Two places P and Q are 92 km apart. A train leaves P for Q and at the same time another train leaves Q for P. Both the trains meet 4 hrs. after they start moving. If the train travelling from P to Q travels 7 km/hr. faster than the other train, find the speed of the two trains.

(1) 15 km/hr, 8 km/hr.

(2) 12 km/hr, 8 km/hr.

(3) 12 km/hr, 9 km/hr

(4) 15 km/hr, 9 km/hr

(5) None of these

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THE TRAIN

Q 20. Two places P and Q are 92 km apart. A train leaves P for Q and at the same time another train leaves Q for P. Both the trains meet 4 hrs. after they start moving. If the train travelling from P to Q travels 7 km/hr. faster than the other train, find the speed of the two trains.

(1) 15 km/hr, 8 km/hr.

(2) 12 km/hr, 8 km/hr.

(3) 12 km/hr, 9 km/hr

(4) 15 km/hr, 9 km/hr

(5) None of these

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THE TRAIN

Q 21. Two trains A and B start from Lucknow and Delhi towards Delhi and Lucknow respectively. After passing each other they take 4 hours and 9 hours to reach Delhi and Lucknow respectively. If the train from Lucknow is moving at 60 km/hr. Then find the speed of the other train.

(1) 40 km/hr.

(2) 30 km/hr.

(3) 35 km/hr.

(4) 50 km/hr.

(5) None of these

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THE TRAIN

Q 21. Two trains A and B start from Lucknow and Delhi towards Delhi and Lucknow respectively. After passing each other they take 4 hours and 9 hours to reach Delhi and Lucknow respectively. If the train from Lucknow is moving at 60 km/hr. Then find the speed of the other train.

(1) 40 km/hr.

(2) 30 km/hr.

(3) 35 km/hr.

(4) 50 km/hr.

(5) None of these

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THE TRAIN

Q 22. Two trains moving from same point same time rspt north and west direction. The speed of one is 5km/hr more than the speed of other. After 2hrs the distance between is 50km. Find the speed of each train?

a) 15,10

b) 20,15

c) 25,20

d) 30,25

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THE TRAIN

Q 22. Two trains moving from same point same time rspt north and west direction. The speed of one is 5km/hr more than the speed of other. After 2hrs the distance between is 50km. Find the speed of each train?

a) 15,10

b) 20,15

c) 25,20

d) 30,25

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THANK YOU

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